NEET UG 2027: Master Biology Chapter Notes

Human Physiology: Respiration, Circulation, Excretion, Locomotion & Coordination

Prepared, Compiled & Authored by Satya Prakash · Calibrated to Rationalized NCERT & NMC Scope

1 Breathing & Exchange of Gases

Respiratory Volumes & Pulmonary Capacities Matrix

All values represent healthy adult human averages measured by spirometer (residual volume cannot be measured by spirometer):

Tidal Volume (TV): 500 mL ($\approx 6000 - 8000$ mL/min for healthy adult)
Inspiratory Reserve Volume (IRV): $2500 - 3000$ mL
Expiratory Reserve Volume (ERV): $1000 - 1100$ mL
Residual Volume (RV): $1100 - 1200$ mL (air remaining after maximal expiration)
Inspiratory Capacity (IC): TV + IRV = $3000 - 3500$ mL
Expiratory Capacity (EC): TV + ERV = $1500 - 1600$ mL
Functional Residual Capacity (FRC): ERV + RV = $2100 - 2300$ mL
Vital Capacity (VC): ERV + TV + IRV = $3500 - 4500$ mL
Total Lung Capacity (TLC): RV + ERV + TV + IRV = VC + RV = $5000 - 6000$ mL

Partial Pressures & Gas Exchange Gradients

Diffusion of gases across the three-layered respiratory membrane (alveolar squamous epithelium, basement membrane, capillary endothelium) occurs along partial pressure gradients:

Gas	Atmospheric Air	Alveoli	Deoxygenated Blood	Oxygenated Blood	Tissues
O_2	159 mmHg	**104** mmHg	40 mmHg	**95** mmHg	40 mmHg
CO_2	0.3 mmHg	**40** mmHg	45 mmHg	**40** mmHg	45 mmHg

Solubility Factor: The solubility of CO_2 is **20–25 times higher** than that of O_2. Hence, the amount of CO_2 that can diffuse across the diffusion membrane per unit difference in partial pressure is much higher. Total membrane thickness is < 1 mm (around 0.2 mm).

Oxygen Dissociation Curve & Gas Transport Mechanisms

1. **Transport of Oxygen:**
- 97% bound to hemoglobin as **Oxyhemoglobin** in RBCs (1 g Hb binds 1.34 mL O_2; each Hb molecule binds max 4 molecules of O_2).
- 3% dissolved in blood plasma.
- **Delivery Yield:** Under normal physiological conditions, every 100 mL of oxygenated blood delivers approximately 5 mL **of** O_2 to the tissues.

2. **Oxygen-Hemoglobin Dissociation Curve (Sigmoid / S-shaped):**
- **Shift to Right (Bohr Effect / O_2 Release at Tissues):** Promoted by:

$$\uparrow pCO_2, \quad \uparrow [H^+] \ (\downarrow \text{pH}), \quad \uparrow \text{Temperature}, \quad \uparrow 2,3\text{-BPG} \tag{1}$$

- **Shift to Left (Loading at Alveoli):** Promoted by $\uparrow pO_2, \downarrow pCO_2, \downarrow [H^+]$ ($\uparrow$ pH), $\downarrow$ Temperature.

3. **Transport of Carbon Dioxide:**
- As **Bicarbonate ions** (HCO_3^-)**: 70**% facilitated by RBC enzyme *carbonic anhydrase* and chloride shift (Hamburger phenomenon).
- As **Carbamino-hemoglobin: 20 – 25**% bound to amino groups of hemoglobin.

- In **Dissolved state** in plasma: **7**%.
- **Delivery Yield:** Every 100 mL of deoxygenated blood delivers approximately 4 mL **of** CO_2 to the alveoli.

NEET Trap: Respiratory Regulation & Clinical Pathology

- **Primary Chemical Sensor:** Chemosensitive area located adjacent to the rhythm center in medulla is **highly sensitive to** CO_2 **and** H^+ **ions**. The role of oxygen in the regulation of respiratory rhythm is **quite insignificant**!
- **Respiratory Centers:** Respiratory rhythm center is in **Medulla oblongata**. **Pneumotaxic center** is in **Pons varolii** (moderates rhythm center and reduces duration of inspiration).
- **Asthma:** Allergic inflammation of bronchi and bronchioles causing wheezing.
- **Emphysema:** Chronic disorder in which **alveolar walls are damaged**, decreasing respiratory surface area; majorly caused by cigarette smoking (α-1-antitrypsin used in treatment).
- **Occupational Respiratory Disorders:** Silicosis, asbestosis; chronic exposure leads to **fibrosis** (proliferation of fibrous tissues) in upper parts of lungs.

2 Body Fluids & Circulation

Blood Composition, Formed Elements & Coagulation Cascade

Blood is a specialized fluid connective tissue consisting of plasma (55%) and formed elements (45%):

- **Plasma:** $90-92\%$ water, $6-8\%$ proteins (Fibrinogen for clotting, Globulins for defense/immunity, Albumins for osmotic balance).
- **Erythrocytes (RBCs):** Biconcave, enucleated in adult mammals, $5.0-5.5\times10^6/\text{mm}^3$. Average lifespan = 120 days; destroyed in the **spleen** (graveyard of RBCs). Hemoglobin count = $12-16\,\text{g}/100\,\text{mL}$.
- **Leukocytes (WBCs):** Nucleated, $6000-8000/\text{mm}^3$.
  - *Granulocytes:* **Neutrophils** ($60-65\%$, phagocytic), **Eosinophils** ($2-3\%$, resist infections, allergic response), **Basophils** ($0.5-1\%$, secrete histamine, serotonin, heparin; mediate inflammation).
  - *Agranulocytes:* **Monocytes** ($6-8\%$, phagocytic), **Lymphocytes** ($20-25\%$, B and T lymphocytes for adaptive immunity).
- **Thrombocytes (Platelets):** Cell fragments produced from **megakaryocytes** in bone marrow ($1.5-3.5\times10^5/\text{mm}^3$).
- **Blood Coagulation Cascade (Enzyme Cascade Theory):**

$$\text{Thromboplastin (Injured Tissue + Platelets)} \xrightarrow{Ca^{2+}} \text{Prothrombinase / Thrombokinase} \tag{2}$$

$$\text{Prothrombin (Inactive)} \xrightarrow[Ca^{2+}]{\text{Thrombokinase}} \text{Thrombin (Active)} \tag{3}$$

$$\text{Fibrinogen (Soluble)} \xrightarrow{\text{Thrombin}} \text{Fibrin (Insoluble Mesh)} \xrightarrow{\text{traps dead elements}} \text{Blood Clot} \tag{4}$$

Calcium (Ca^{2+}) plays an indispensable cofactor role in almost all clotting steps!

Cardiac Cycle, Hemodynamics & Electrocardiogram (ECG)

1. **Cardiac Cycle Timing** ($72\,\textbf{beats/min} \implies \textbf{Duration} = 0.8\,\text{sec}$):

Atrial Systole: 0.1 sec (increases ventricular filling by 30%)
Ventricular Systole: 0.3 sec (isovolumetric contraction $\rightarrow$ ejection period)
Joint Diastole: 0.4 sec (passive filling of atria and ventricles)
Stroke Volume (SV): EDV − ESV = 120 mL − 50 mL = **70** mL
Cardiac Output (CO): Stroke Volume × Heart Rate = 70 mL × 72/min ≈ **5040** mL/min ≈ **5** L/min

2. **Heart Sounds:**

- **First Heart Sound (LUBB):** Closure of **atrioventricular valves** (Tricuspid and Bicuspid) at the

onset of ventricular systole; low pitched, longer duration.
- **Second Heart Sound (DUBB):** Closure of **semilunar valves** at the onset of ventricular diastole; high pitched, shorter duration.

3. **Standard Electrocardiogram (ECG) Deflections:**
- **P-wave:** Depolarization of atria (leads to atrial contraction).
- **QRS Complex:** Depolarization of ventricles (marks onset of ventricular contraction; counting QRS complexes per unit time gives heart rate).
- **T-wave:** Repolarization of ventricles (return to resting state; end of T-wave marks end of systole).

Double Circulation & Hepatic Portal System

- **Pulmonary Circulation:** Deoxygenated blood from Right Ventricle $\xrightarrow{\text{Pulmonary Artery}}$ Lungs $\xrightarrow{\text{Pulmonary Veins}}$ Left Atrium.
- **Systemic Circulation:** Oxygenated blood from Left Ventricle $\xrightarrow{\text{Aorta}}$ Body Tissues $\xrightarrow{\text{Vena Cava}}$ Right Atrium.
- **Hepatic Portal System:** Unique vascular connection carrying blood from the **digestive tract to the liver** via the Hepatic Portal Vein before reaching systemic circulation.
- **Coronary Circulation:** Exclusive network of blood vessels supplying muscular walls of the heart.

3 Excretory Products & Their Elimination

Modes of Nitrogenous Excretion & Nephron Anatomy

- **Ammonotelism:** Excretion of toxic ammonia requiring large volumes of water; diffuses directly across body surfaces or gill surfaces as ammonium ions (e.g., bony fishes, aquatic amphibians, aquatic insects). Kidneys play no significant role in ammonia removal.
- **Ureotelism:** Urea synthesized in the liver via Ornithine cycle; requires moderate water (e.g., mammals, terrestrial amphibians, marine cartilaginous fishes).
- **Uricotelism:** Uric acid excreted as pellet/paste with minimal water loss; least toxic (e.g., reptiles, birds, land snails, insects).
- **Nephron Types:**
  - **Cortical Nephrons (85%):** Henle's loop is short and extends only very little into medulla; vasa recta absent or highly reduced.
  - **Juxtamedullary Nephrons (15%):** Henle's loop is extremely long and runs deep into the inner medulla; well-developed vasa recta; **essential for concentrating urine**.

Renal Physiology: GFR & Counter-Current Mechanism

1. **Glomerular Filtration Rate (GFR):**

$$\text{Renal Blood Flow} \approx 1100 - 1200\,\text{mL/min} \quad (\approx 20 - 25\% \text{ of Cardiac Output}) \tag{5}$$

$$\textbf{GFR} = \textbf{125}\,\text{mL/min} = \textbf{180}\,\text{litres/day} \implies \text{Urine output} \approx \textbf{1.5}\,\text{litres/day (99\% reabsorbed!)} \tag{6}$$

2. **Net Filtration Pressure (NFP):**

$$\text{NFP} = \text{GBHP} - (\text{BCOP} + \text{CHP}) = 60 - (32 + 18) = \textbf{10}\,\text{mmHg} \tag{7}$$

3. **Counter-Current Multiplier & Exchanger:**
- Operated between **Loop of Henle** and **Vasa Recta**.
- Flow of filtrate in two limbs of Henle's loop is in opposite directions; flow of blood in two limbs of vasa recta is also in opposite directions.
- **Descending Limb of Henle:** Permeable to water, almost impermeable to electrolytes $\implies$ filtrate becomes concentrated as it moves down ($300 \rightarrow 1200\,\text{mOsmol/L}$).

- **Ascending Limb of Henle:** Impermeable to water, actively/passively transports electrolytes (NaCl) into medullary interstitium $\implies$ filtrate becomes diluted as it ascends (1200 $\rightarrow$ 200 mOsmol/L).
- Creates an osmolarity gradient in medullary interstitium: 300 mOsmol/L **in cortex** $\rightarrow$ 1200 mOsmol/L **in inner medulla**.
- Human kidneys can produce urine nearly **4 times more concentrated** than the initial filtrate (1200 mOsmol/L vs 300 mOsmol/L).

Hormonal Regulation of Kidney Function

1. **RAAS (Renin-Angiotensin-Aldosterone System):**

$$\downarrow \text{GFR / Blood Pressure} \xrightarrow{\text{detected by JGA}} \text{Juxtaglomerular cells release } \mathbf{Renin} \tag{8}$$

$$\text{Angiotensinogen (Liver)} \xrightarrow{\text{Renin}} \text{Angiotensin I} \xrightarrow{\text{ACE}} \text{Angiotensin II (Potent Vasoconstrictor)} \tag{9}$$

   Angiotensin II increases GFR directly by arteriolar constriction AND stimulates **adrenal cortex** to secrete **Aldosterone** $\implies$ promotes active reabsorption of Na^+ and water from DCT $\implies \uparrow$ Blood Volume and BP.
2. **Antidiuretic Hormone (ADH / Vasopressin):** Released by posterior pituitary when hypothalamic osmoreceptors detect $\uparrow$ blood osmolarity (dehydration); makes DCT and Collecting Duct permeable to water via aquaporins, preventing diuresis.
3. **Atrial Natriuretic Factor (ANF):** Released by atrial walls of heart in response to $\uparrow$ blood flow/pressure; causes **vasodilation** and promotes natriuresis (Na^+ excretion). Acts as a negative feedback check on RAAS!

4 Locomotion & Movement: Musculoskeletal Mechanics

Sarcomere Architecture & Sliding Filament Theory

A sarcomere is the basic functional contractile unit of a muscle fiber, bounded between **two successive Z-lines**:

- **A-Band (Anisotropic):** Contains both thick (myosin) and thin (actin) filaments; length **remains completely constant** during contraction.
- **I-Band (Isotropic):** Contains only thin actin filaments; bisected by elastic **Z-line**; **shortens** during contraction.
- **H-Zone:** Central region of A-band where thick filaments are not overlapped by thin filaments; bisected by fibrous **M-line**; **disappears** during maximal contraction.

Myofilament Biochemistry:

- **Thin Filament:** Composed of 2 helically wound polymeric **F-actin** filaments, 2 filamentous **Tropomyosin** running along F-actin grooves, and complex **Troponin** triplets (TnT, TnI, TnC). **Troponin-C (TnC)** binds Ca^{2+}, unmasking active myosin-binding sites on actin.
- **Thick Filament:** Composed of polymerized **Myosin** monomer units (**Meromyosins**):
  - **Heavy Meromyosin (HMM):** Globular head and short arm; projects outwards as cross-bridge; contains actin-binding site and active **ATPase enzyme site**.
  - **Light Meromyosin (LMM):** Tail portion.

Contraction Cycle: Action potential depolarizes sarcolemma $\rightarrow$ Sarcoplasmic reticulum releases Ca^{2+} $\rightarrow$ Ca^{2+} binds to Troponin-C $\rightarrow$ Tropomyosin shifts $\rightarrow$ Myosin head hydrolyzes ATP (ADP + P_i) and binds actin forming cross-bridge $\rightarrow$ **Power Stroke** pulls thin filaments towards center of A-band $\rightarrow$ New ATP binds myosin head, causing detachment.

Human Skeletal Framework (206 Bones) & Articulations

- **Axial Skeleton (80 Bones):**
  - **Skull (22 bones):** Cranial (8) + Facial (14). Associated: Hyoid (1) + Ear Ossicles ($2 \times 3 = 6$:

Malleus, Incus, Stapes).
  - **Vertebral Column (26 in adult):** Cervical (7, C_1 = Atlas, C_2 = Axis), Thoracic (12), Lumbar (5), Sacral (1 fused), Coccygeal (1 fused).
  - **Thoracic Cage:** Sternum (1) + Ribs (12 pairs = 24). True ribs ($1^{st} - 7^{th}$ pair, vertebrosternal), False ribs ($8^{th}, 9^{th}, 10^{th}$ pair, vertebrochondral), Floating ribs ($11^{th}, 12^{th}$ pair, vertebral).
- **Appendicular Skeleton (126 Bones):**
  - **Forelimb ($30 \times 2 = 60$):** Humerus (1), Radius (1), Ulna (1), Carpals (8 wrist), Metacarpals (5 palm), Phalanges (14 digits, formula 2,3,3,3,3).
  - **Pectoral Girdle ($2 \times 2 = 4$):** Clavicle (collar bone) + Scapula (has acromion process, spine, and glenoid cavity for head of humerus).
  - **Hindlimb ($30 \times 2 = 60$):** Femur (thigh, longest/strongest), Patella (kneecap, sesamoid bone), Tibia (1), Fibula (1), Tarsals (7 ankle), Metatarsals (5), Phalanges (14).
  - **Pelvic Girdle (2 coxal bones):** Each formed by fusion of Ilium, Ischium, and Pubis; meet ventrally at pubic symphysis (fibrous cartilage); cavity = **Acetabulum** (articulates with femur).
- **Joint Classifications:**
  - **Fibrous (Synarthroses):** Immovable (sutures between flat skull bones).
  - **Cartilaginous (Amphiarthroses):** Limited movement (pubic symphysis, between adjacent vertebrae).
  - **Synovial (Diarthroses):** Fluid-filled synovial cavity; freely movable.
    - *Ball and Socket:* Shoulder (glenoid-humerus) and Hip (acetabulum-femur).
    - *Hinge Joint:* Knee joint, Elbow joint, Interphalangeal joints.
    - *Pivot Joint:* Between Atlas and Axis ($C_1 - C_2$).
    - *Gliding Joint:* Between adjacent Carpals.
    - *Saddle Joint:* Between Carpal and Metacarpal of human thumb.

5 Neural Control & Chemical Coordination

Nerve Impulse: Resting, Depolarization & Repolarization

Resting Membrane Potential: -70 mV (Inside negative; permeable to K^+, impermeable to Na^+)

Sodium-Potassium Pump: Transports **3** Na^+ out for **2** K^+ in (Consumes 1 ATP)

Action Potential: $+30$ mV (Voltage-gated Na^+ channels open $\Longrightarrow$ Na^+ influx)

Repolarization: Na^+ channels close, voltage-gated K^+ channels open $\Longrightarrow$ K^+ efflux

Synapse: AP at terminal $\rightarrow$ Ca^{2+} influx $\rightarrow$ ACh exocytosis $\rightarrow$ receptors

Endocrine Master Table: Glands, Hormones, Target & Disorders

Endocrine Gland	Hormone	Nature	Major Action & Pathology
Anterior Pituitary	Growth Hormone (GH)	Peptide	Childhood excess: **Gigantism**; adult excess: **Acromegaly**; deficiency: **Pituitary Dwarfism**.
Posterior Pituitary	Oxytocin, Vasopressin	Peptide	Oxytocin: uterine contraction & milk ejection. Vasopressin: water reabsorption; deficiency $\Longrightarrow$ **Diabetes Insipidus**.
Thyroid	Thyroxine (T_4, T_3)	Iodinated	BMR regulation. Pregnancy deficiency $\Longrightarrow$ **Cretinism**; adult: **Myxoedema**; excess: **Graves' disease**.
Thyroid / C-cells	Thyrocalcitonin (TCT)	Peptide	**Hypocalcemic**; lowers blood Ca^{2+} via bone deposition.
Parathyroid	Parathormone (PTH)	Peptide	**Hypercalcemic**; bone resorption; antagonist to TCT.
Adrenal Cortex	Aldosterone, Cortisol	Steroid	Na^+ retention. Hyposecretion $\Longrightarrow$ **Addison's disease**; hypersecretion $\Longrightarrow$ **Cushing's syndrome**.
Adrenal Medulla	Adrenaline	Catecholamine	Emergency / Fight-or-Flight; $\uparrow$ heart rate, glycogenolysis.
Pancreas (β-cells)	Insulin	Peptide	Hypoglycemic; promotes cellular glucose uptake; deficiency $\Longrightarrow$ **Diabetes Mellitus**.
Pancreas (α-cells)	Glucagon	Peptide	Hyperglycemic; stimulates glycogenolysis; antagonist to insulin.

Mechanism of Hormone Action: Peptide vs Steroid Pathway

- **Peptide / Protein / Amine Hormones (Water-Soluble):** Insulin, glucagon, pituitary hormones, epinephrine. Cannot cross lipid bilayer; bind to **extracellular membrane-bound receptors** $\Longrightarrow$ generate second messengers (**cyclic AMP,** IP_3, Ca^{2+}) $\Longrightarrow$ cascade modifies cellular biochemical metabolism.
- **Steroid & Iodothyronine Hormones (Lipid-Soluble):** Estrogen, progesterone, testosterone, cortisol, thyroxine (T_3/T_4). Cross cell membrane freely; bind to **intracellular / nuclear receptors** $\Longrightarrow$ hormone-receptor complex interacts with genome/chromosome to regulate gene expression and protein synthesis.